

Yuntian Liu

ytlou0102.github.io

EDUCATION

Nanyang Technological University - Singapore, Singapore

09/2024 – 06/2026

Master of Science in Electronics

Wuhan University - Wuhan, China

09/2020 – 06/2024

Bachelor of Electronic Information Engineering

RESEARCH INTERESTS

- Energy-Efficient Edge Accelerators | Compilers & Scheduling for Heterogeneous Systems | HW/SW Co-Design

PUBLICATIONS

- R. Harish, V. P. Nambiar, **Y. Liu**, et al., "A CGRA with SIMD and AGU", manuscript completed, to be submitted to *International Conference on Computer-Aided Design (ICCAD)*, 2027.
- **Y. Liu**, et al., "Tree-Inspired Techniques for Diffusion Models", manuscript completed, to be submitted to *International Symposium on Computer Architecture (ISCA)*, 2027.
- **Y. Liu**, et al., "Distance-Free Farthest Point Sampling Accelerator for Point-Cloud Networks", in preparation for *IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II)*, 2026.
- **Y. Liu**, Y. S. Chong, B. Lienhard, M. Fan, W. L. Goh, V. P. Nambiar, and A. T. Do, "Qubit-State Discrimination using Neural Networks with Rapid and Energy-Efficient Compute Arrays," *IEEE Int. Symp. Circuits Syst. (ISCAS)*, 2025.

RESEARCH EXPERIENCE

Custom CGRA Instruction Set and Compiler Integration

*Research Intern & Master's Thesis Project, Institute of Microelectronics (IME), A*STAR | 09/2025 – 01/2026*

- Defined a SIMD instruction set for CGRA routers and PEs, improving energy efficiency by up to 3.96×.
- Built the assembler and an AGU-aware loop scheduler, improving PE utilization by 2×.
- Implemented in 12nm FinFET through post-layout: 1 GHz, 1.31 TOPS/W peak, 3.4× over state of the art.

Tree-Inspired Techniques for Diffusion Models

supervised by Dr. [Changchun Zhou](#) | 04/2026 – present

- Proposed Morton-order block dropout, removing 95.1% of SDXL denoiser FLOPs (15.9× speedup over an L40S GPU).
- Designed a sparse diffusion SoC with 1.56–2.67× higher energy efficiency than SOTA accelerators at 0.63% area.
- Implemented in SystemVerilog with Vivado synthesis and FPGA validation.

Distance-Free Farthest Point Sampling Accelerator for Point-Cloud Networks

supervised by Dr. [Changchun Zhou](#) | 07/2026 – present

- Proposed a counting-based FPS accelerator that replaces distance search with per-block occupancy counting.
- Removed the serial arg-max for a parallel streaming schedule, 26× faster than exact FPS at 64k points.
- Implemented as an algorithm model; TCAS-II brief in preparation.

Scheduling Algorithm for Heterogeneous Systems

supervised by Dr. [Changchun Zhou](#) | 09/2025 – 05/2026

- Built a dependency-aware scheduler for the Nebula NPU (ISSCC 2025) that plans data transfers for DAGs.
- Allocated on-/off-chip storage and generated data-transfer instructions on the fly; deployed on the 28nm Nebula chip.

Hardware-Friendly Neural Network for Qubit State Determination

Final Year Project | supervised by Prof. [Wang Ling Goh](#) and Dr. [Anh-Tuan Do](#) | 01/2024 – 10/2024

- Compressed the readout FCNN by 51.3× via narrower layers, ReLU and INT8 QAT at 90.86% fidelity (0.04% drop).
- Designed an output-stationary PE array with 63 ns latency and $\geq 1.99\times$ higher energy efficiency.
- Implemented in 28nm at 1 GHz (1.04 mm², 1.8 TOPS/W) and mapped to FPGA.

SKILLS & TRAITS

- Flows: Hardware–Software Co-Design, Custom ISA & Assembler Development, Task Scheduling, NN Quantization
- Tools & Platforms: RISC-V, gem5, MLIR, FPGA | Languages: Verilog, Python, C, C++